

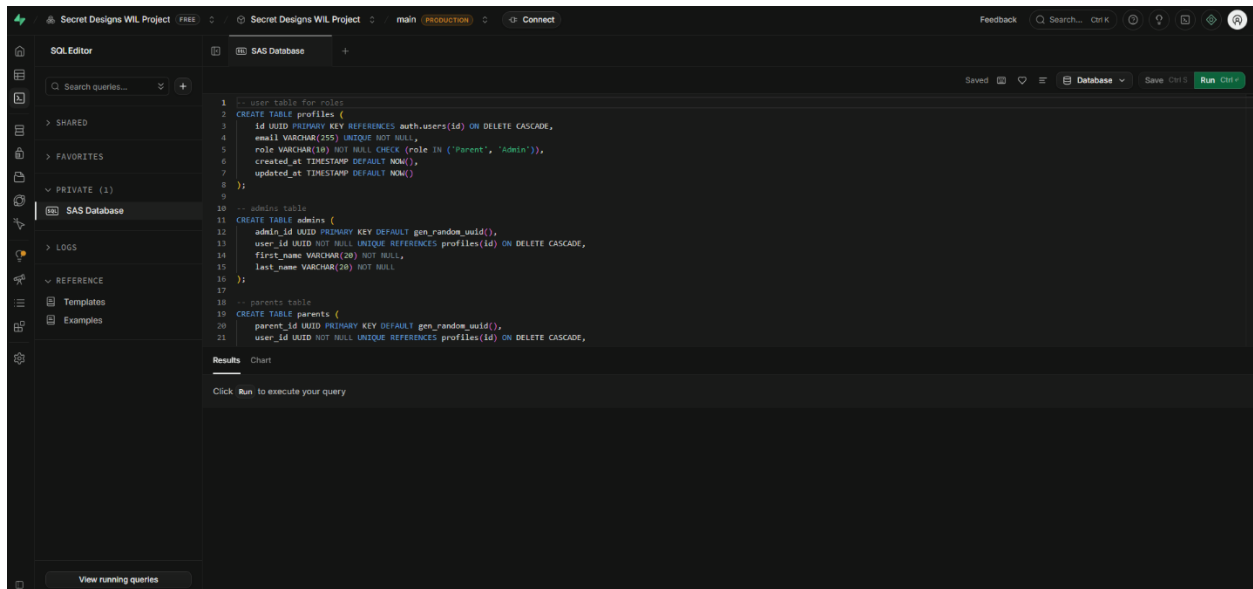
Name: Lebohang Molise

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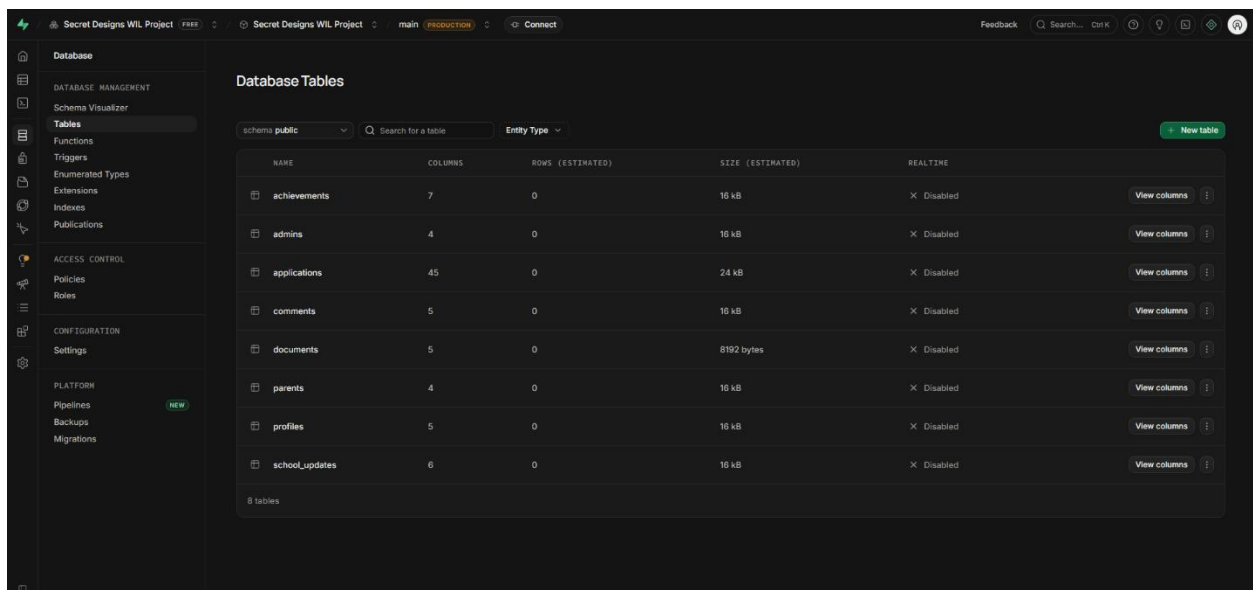
Individual Contributions and Reflection

1. Responsible for creating database tables in Supabase

- Ensured the database tables match with the requirements
- Plotted out the database design before implementing it on Supabase

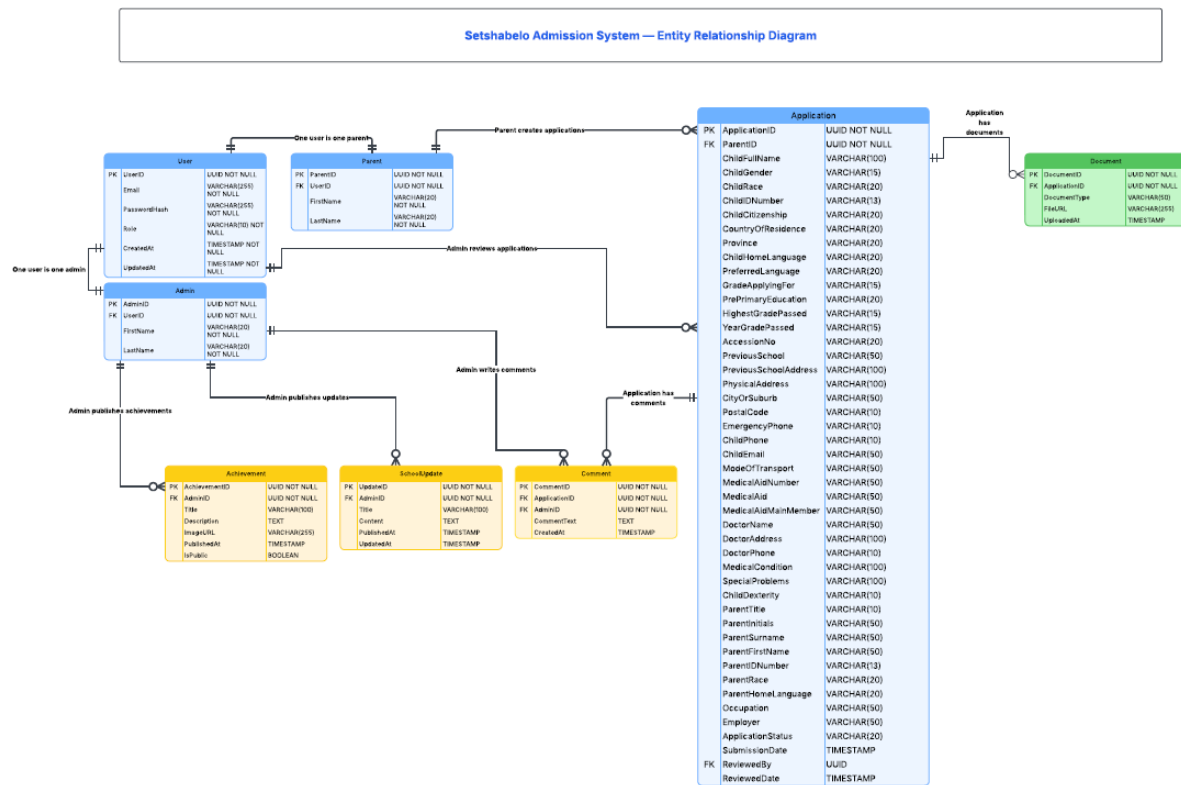


Supabase SQL Editor where I created the tables



Database tables created on Supabase

I was also responsible for redesigning the ERD Diagram to fit all the fields that were not included on the Application Entity



Updated ERD Diagram

2. Responsible for Unit 3 SRS Documentation

- Ensured appropriate format for the SRS
- Ensured all the information on the SRS lines up with all the work done so far

UNIT 3 – System Architecture & Design

System Architecture & Design

23. Main System Components

Component	Purpose
Flutter Mobile Application	Provides mobile functionality for the intended mobile users
ASP.NET Web Application	Provides web functionality for the intended web or administrative users
Supabase Authentication	Manages authentication where required
Supabase Database	Stores information used by both applications
Supabase Storage	Stores files or documents where required

24. System User and Roles

User / Role	Uses	Main purpose
Parent / Guardian	Flutter	Create an account, submit learner applications, upload documents, view application status, view school updates and achievements
School Admins	ASP.NET	Log in, view all applications, accept or reject applications, leave comments, publish school updates and achievements

25. Component Responsibilities

Component	Main Responsibilities
Flutter	Allow parents to register and log in, complete the multi-step application form, upload supporting documents, view application status, view school updates, and view achievements.
ASP.NET	Allow admins to log in, view all submitted applications, accept or reject applications, add comments, publish school updates, and publish achievements.

Supabase Authentication	Verify user identity and differentiate between Parent and Admin roles.
Supabase Database	Store user profiles, parent and admin records, applications, documents metadata, comments, school updates, and achievements.
Supabase Storage	Store uploaded files (birth certificates, parent IDs, etc) securely.

+ 26. Main Data Flows

From	To	Information Exchanged
Parent	Flutter	Login credentials, application form data, document uploads
Flutter	Supabase Auth	Authentication requests (sign up, sign in)
Flutter	Supabase Database	Application data, status requests
Flutter	Supabase Storage	Uploaded documents
Supabase Database	Flutter	Application status, school updates, achievements
Supabase Storage	Flutter	Document URLs for viewing
Admin	ASP.NET	Login credentials, review decisions, published content
ASP.NET	Supabase Auth	Authentication requests (admin login)
ASP.NET	Supabase Database	Application retrieval, status updates, content publishing
Supabase Database	ASP.NET	All application data, comments, published content
Admin	ASP.NET	Accept/reject decisions, comments, new updates

28. UML & Database Design

Use Case Diagram

Identifying Actors

Actor	What does this actor need to do?
Parent / Guardian	Register an account, log in, submit learner applications, upload supporting documents, view application status, view school updates, view school achievements
School Admin	Log in, view all submitted applications, review applications, accept or reject applications, leave comments for parents, publish school updates, publish school achievements

29. Identifying Use Cases

Actor	Possible Use Case
Parent / Guardian	Register
Parent / Guardian	Login
Parent / Guardian	Submit Application
Parent / Guardian	Upload Documents
Parent / Guardian	View Application Status
Parent / Guardian	View Application Comments
Parent / Guardian	View School Updates
Parent / Guardian	View Achievements
School Admin	Login
School Admin	View All Applications
School Admin	Review Application
School Admin	Accept Application
School Admin	Reject Application
School Admin	Add Comment
School Admin	Publish School Update
School Admin	Publish Achievement

User Interface Design

33. Required Flutter Screens

Flutter Screen	User	Purpose	Related Requirement
Splash / Login	Parent	Authenticate user	FR1, FR7
Register	Parent	Create a new parent account	FR1
Parent Home	Parent	Main dashboard after login	FR1, FR2
Child Info (Step 1 of form)	Parent	Capture child's personal details	FR2
Address (Step 2 of form)	Parent	Capture home address and contact details	FR2
Medical Info (Step 3 of form)	Parent	Capture medical aid and doctor details	FR2
Parent Info (Step 4 of form)	Parent	Capture parent details	FR2
Documents (Step 5 of form)	Parent	Upload supporting documents	FR2, IR2
Review & Submit (Step 6 of form)	Parent	Review all info and submit	FR2
My Applications	Parent	View all submitted applications and their status	FR3
Application Detail	Parent	View full detail of a single application + comments	FR3
School Updates	Parent	View announcements from the school	FR5
Achievements	Parent	View school achievements	FR6

34. Required ASP.NET Screens

ASP.NET Page	User	Purpose	Related Requirement
Login	Admin	Authenticate admin	FR7, IR4
Admin Dashboard	Admin	Overview of applications and quick actions	FR4
Application List	Admin	View all submitted applications	FR4

Navigation Between Screens

35. Flutter Mobile Navigation Flow

Starting Screen	User Action	Next Screen
Splash / Login	Tap "Register"	Register
Register	Tap "Login"	Splash / Login
Splash / Login	Successful login (Parent)	Parent Home
Parent Home	Tap "New Application"	Child Info (Step 1)
Child Info	Tap "Save"	Address (Step 2)
Address	Tap "Save"	Medical Info (Step 3)
Medical Info	Tap "Save"	Parent Info (Step 4)
Parent Info	Tap "Save"	Documents (Step 5)
Documents	Tap "Save"	Review & Submit (Step 6)
Review & Submit (Step 6)	Tap "Apply"	Success Screen
Success Screen	Tap "Back to dashboard"	Parent Home
Parent Home	Tap "My Applications"	My Applications
My Applications	Tap an application	Application Detail
Parent Home	Tap "School Updates"	School Updates
Any screen	Tap "Logout"	Splash / Login

36. ASP.NET Web Navigation Flow

Starting Screen	User Action	Next Screen
Login	Tap "Sign In"	Admin Dashboard
Admin Dashboard	Tap "View All Applications"	Application List
Application List	Tap an application	Application Detail
Application Detail	Tap "Accept" or "Reject"	Application Detail (updated status)
Application Detail	Tap "Back"	Application List
Admin Dashboard	Tap "Publish Update"	Publish Update
Publish Update	Tap "Publish"	Admin Dashboard (with success)
Admin Dashboard	Tap "Publish Achievement"	Publish Achievement

37. Mobile & Web Integration

Flutter Action	Supabase Data	ASP.NET Action
Parent submits application	Application row created	Admin views application in list
Parent uploads documents	Document rows created and files in Storage	Admin views documents from Application Detail
Parent views status	Fetches application_status from Application	Admin updates status to "Accepted" or "Rejected"
Parent views updates	Fetches SchoolUpdate rows	Admin publishes new School Update
Parent views achievements	Fetches Achievement rows	Admin publishes new Achievement

38. Stakeholder Feedback

Item	Detail
Date of feedback	16 September 2026
Designs shown	All Flutter Wireframes
Stakeholder comments	The form is easy to follow, and it includes the important fields. The design is appropriate and fits with the school. The screens look good
Changes requested	None
Changes made	None
Evidence available	Signed Feedback Sheet

39. Updated System Design

Design Artifact	Change Made	Reason
Architecture	No change	Architecture confirmed
Use Case Diagram	No change	Use cases confirmed
Sequence Diagram	No change	Processes confirmed
ERD	No change	ERD already supports optional fields
Flutter Interface	No change	Stakeholder feedback
ASP.NET Interface	No change	Interfaces confirmed